Task 2: Data Cleaning & Preprocessing

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1. Introduction

The purpose of this task is to clean and prepare a dataset for future analysis, visualization, and prediction.

Data cleaning is an important step in data science because missing values, duplicate records, incorrect data types, and inconsistent values can affect the accuracy of analysis.

For this task, I used an e-commerce order dataset provided by DecodeLabs. The dataset contains information about customer orders, products, payment methods, order statuses, coupon codes, referral sources, and total prices.

2. Dataset Overview

The dataset contains:

Item	Value
Number of rows	1200
Number of columns	14

Table 1: Dataset Overview Table

Each row represents one customer order. Each column contains information related to the order, customer, product, payment, shipping, or price.

3. Column Description

Column Name	Description	Data Type
OrderID	Unique ID for each order	Categorical/Alpha-Numerical
Date	Date when the order was placed	Date
CustomerID	Unique ID for each customer	Categorical/Alpha-Numerical
Product	Product purchased by the customer	Categorical/Text
Quantity	Number of units purchased	Numerical
UnitPrice	Price of one unit of the product	Numerical
Shipping Address	Shipping address of the customer	Alpha-Numerical
PaymentMethod	Payment method used for the order	Categorical/Text
OrderStatus	Status of the order	Categorical/Text
TrackingNumber	Tracking number for shipment	Alpha-Numerical
ItemsInCart	Number of items in the customer's cart	Numerical
CouponCode	Coupon code used for the order, if any	Alpha-Numerical
ReferralSource	Source that referred the customer	Categorical/Text
TotalPrice	Total price of the order	Numerical

Table 2: Category Table

4. Missing Values and Duplicates

The dataset was checked for missing values in each column.

Column Name	Missing Values
OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0

Table 3: Missing Values Table

The only column with missing values is CouponCode, which has 309 missing. This is reasonable since not every customer uses a coupon code when placing an order. TO handle this issue, the missing values in the CouponCode column should be replaced with “No Coupon”. This keeps information meaningful instead of deleting rows.

The dataset was also checked for duplicate rows. None were found, meaning there were no repeated order records that needed to be removed.

5. Category Consistency

The categorical columns were checked for consistent values.

Column Name	**Unique Values**
Product	Monitor, Phone, Tablet, Chair, Printer, Laptop, Desk
PaymentMethod	Debit Card, Online, Credit Card, Gift Card, Cash
OrderStatus	Shipped, Cancelled, Returned, Delivered, Pending
CouponCode	SAVE10, FREESHIP, WINTER15, No Coupon
ReferralSource	Instagram, Referral, Email, Facebook, Google

Table 4: Category Label Table

The category values are consistent. There are no obvious spelling mistakes or duplicated categories with different formatting.

6. Numerical Value Check

The numerical columns were checked to make sure their values are reasonable.

Column Name	Minimum Value	Maximum Value
Quantity	1	5
UnitPrice	11.39	699.93
ItemsInCart	1	10
TotalPrice	11.39	3456.40

Table 5: Numerical Value Check Table

The values are reasonable for an order dataset. Quantity and ItemsInCart are positive values, UnitPrice is positive, and TotalPrice is also positive.

7. Total Price Validation

The TotalPrice column was checked using the formula:

$$\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$$

The TotalPrice values matched the expected calculation. This means the TotalPrice column is valid and does not need correction.

8. Cleaning Actions Taken

Cleaning Step	Action Taken
Missing values	Replaced missing CouponCode values with “No Coupon”
Duplicate rows	Checked; none found
Date format	Date column should be kept in date format
Category consistency	Checked; values are consistent
Numerical values	Checked; values are reasonable
Total price	Checked; values are correct

Table 6: Action Taken Table

9. Task 2 Summary

In this task, the e-commerce order dataset was checked and prepared for future data analysis.

The dataset contains 1200 rows and 14 columns. The main cleaning issue was found in the CouponCode column, which had 309 missing values. Since missing coupon values mean that no coupon was used, these missing values should be replaced with "No Coupon".

There were no duplicate rows in the dataset. The categorical values were consistent, and the numerical columns had reasonable values. The TotalPrice column was also checked and confirmed to match Quantity multiplied by UnitPrice.

Overall, the dataset is now ready for exploratory data analysis, visualization, and further data science tasks.